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Review Article

Asymptomatic cholecystitis and its controversial issues in biliary surgery

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ABSTRACT

Gallbladder stones are present in 10–20 % of the population of Western countries, and the incidence rate has almost tripled over the past 30 years. Asymptomatic cholecystitis is more common than it is mentioned in the literature. Approximately 80 % of the GI have no clinical manifestations. In most cases, the disease will remain asymptomatic throughout life.

The problem of asymptomatic cholecystitis is often hushed up and ignored. Surgeons try to avoid examining this pathology. Works on asymptomatic cholecystitis are not only few, but also scattered. At the same time, there is no uniformity in the literature. Since the tactics of treating such patients still does not have an unambiguous approach. Many surgeons doubt the decision on surgical treatment, a smaller part chooses cholecystectomy at the same time, without substantiating the indications for it. That is why most cases of postcholecystectomy syndrome occur in patients operated on for asymptomatic cholecystitis.

The unwillingness to discover the secret of stone formation and asymptomatic cholecystitis in humans when concretions in the gallbladder are detected without reference to surgery, to indicate the trajectory of the examination, leads to a number of tactical errors, preserving the risk of PES, and erasing practical recommendations for the management of such patients. The solution to this issue lies in a thorough study of each stage of the tactical search in patients with asymptomatic cholecystitis, which will determine a new strategy in biliary surgery.

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1. Annotation

Gallbladder stones are a common public health problem, especially in developed countries. In Russia, the annual incidence of cholelithiasis averages 5–6 people per 1000 population. In Western countries, 10–20 % of people have gallbladder stones, and in the past 30 years, the incidence rate has nearly tripled (also because of better diagnostics). It has been determined that the disease is more common in certain age and gender groups. Women are five to six times more likely than men to have cholelithiasis. The number of patients significantly increases with age: after 70 years of age it reaches 30 % and more, i.e. pathology has become one of the serious geriatric problems.[1] The prevalence of asymptomatic calculous cholecystitis is higher than what is documented in the literature. About 80 % of cases of cholelithiasis do not exhibit any clinical symptoms. It is reasonable to assume that the majority of the time the disease will remain asymptomatic throughout life based on the analysis of the disease's course. Clinical manifestations develop in 1–4% of patients per year; in 20 years from the time of diagnosis, symptoms appear in 20 % of patients.[2,3] With the increased emphasis on medical screening and the development of diagnostic imaging techniques such as magnetic resonance, endoscopic

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retrograde cholangiopancreatography and endoscopic ultrasound, more and more asymptomatic cases of cholelithiasis are being identified.

However, the issue of asymptomatic cholecystitis is frequently downplayed or disregarded. Surgeons make an effort not to look into this pathology. Since the tactics of treatment of such patients still has no unambiguous approach. Most surgeons are hesitant to make a decision on surgical treatment, a smaller part chooses cholecystectomy in this case without substantiating the indications for it. Therefore, most cases of postcholecystectomy syndrome are precisely in patients operated on for asymptomatic cholecystitis.

2. Introduction

The management course for these patients is still unclear in clinical recommendations. On the one hand, they write that it is necessary to carefully approach the examination of such patients. On the other hand, in the conclusion of recommendations it is written that works on research of the problem of management of patients with asymptomatic cholecystitis are in priority in the scientific community dealing with cholelithiasis. There is little debate among clinicians about the treatment of symptomatic choledocholithiasis, which can sometimes lead to life-threatening conditions. However, the treatment of asymptomatic choledocholithiasis is becoming controversial. Any treatment is associated with a certain frequency of adverse events, and treatment with endoscopic retrograde cholangiopancreatography is no exception.

There have been eight relevant reports and four national guidelines from the National Institute for Health and Care Excellence in the UK (NICE), the European Society for Gastrointestinal Endoscopy (ESGE), the British Society of Gastroenterology (BSG) and the European Association for the Study of the Liver (EASL) on the management of asymptomatic choledocholithiasis.[4] Among them, three of them showed the expediency of a wait-and-see strategy due to spontaneous removal of stones. Thus according to EASL guidelines: treatment of patients with asymptomatic gallstones is not recommended (weak recommendation based on very low quality evidence).[5] Since some patients may have persistence or recurrence of symptoms after cholecystectomy. Indeed, 10–40 % of patients have persistent complaints after cholecystectomy, sometimes occurring after a short asymptomatic period.[6] Meanwhile, three of the four national guidelines (NICE, ESGE and BSG) support the removal of stones from asymptomatic common bile duct with some form of procedure, even though there is little data on the natural course of asymptomatic choledocholithiasis and no randomized controlled trial between intervention and observation. As a result, we are now faced with a dilemma.

What is known, however, is that every study gives top priority to analyzing the cost-effectiveness of laparoscopic cholecystectomy while accounting for the clinical course of the disease in patients with asymptomatic or asymptomatic gallstones; investigations are also carried out on the risk of complications and biliary colic in patients with bile sludge in the gallbladder or asymptomatic biliary tract stones.

3. Methods

In case of rejection of literature, we used the rad high-tech databases: PubMed/Medline, semanticscholar.org, scholar.google.com. All the articles obtained as a result of the search were translated to the Rayyan platform. The Rayyan platform has allowed us to significantly simplify the selection of necessary publications and eliminate repetitive research. Three independent authors read all the abstracts and applied the inclusion and exclusion criteria, which were taken from the criteria of the Oxford Center (Table 1).

To search for literary sources, we used the following words and phrases: “asymptomatic cholecystitis”, “latent cholecystitis”, “asymptomatic cholelithiasis”, “calculous cholecystitis”, “cholecystolithiasis”, “choledocholithiasis”, “cholecystectomy”. The selected publications were classified according to the level of evidence from I to V degree and recommendations from A to D for evidence-based medicine. After that, they reviewed the selected studies in detail in order to draw the conclusions presented in our review. In addition, for a clinical expert of various kinds, we also adopted Shukala Jadad, she called for a better expression of the methodological quality of each individual study. For use in our analysis of abstracts and full-text articles, we conducted a thorough selection process based on the content of studies on asymptomatic cholecystitis and modern views on the management of such patients from the perspective of the latest technical diagnostic and treatment capabilities, including surgical ones. The annual evaluation of the Guidelines for Research and Evaluation II (AGREED II) of December 2017 included a full-scale, comprehensive and immediate analysis, all studies analyzed. In addition, the Tokyo Manual (2018) was partially involved in the controversial studies in order to better clarify minor or unaccounted-for points.[28]

4. Discussion

So, when considering the course of asymptomatic cholecystitis, there is never a clear clinical picture. All that doctors grasp for are two facts: recurrent pain in the right subcostal region and the presence of concretions in the gallbladder. And the latter are often a finding in abdominal screening examinations. The patient's doubts about the need for surgery are shattered by the argument about the possible complications of long-term stone retention. And difficulties in interpretation arise when the usual manifestation of common bile duct stones is the same acute biliary colic due to its distension due to partial or complete obstruction as cholelithiasis. This pain is extremely difficult to distinguish from the pain of gallstones. Unlike gallstones, common bile duct stones are symptom-free only in 5–12 % of cases. Small stones obturating the sphincter of Oddi can also cause distal obstruction and acute pancreatitis.[7]

On the other hand, even when endoscopists fully inform patients about the risks before the procedure, the occurrence of complications after endoscopic retrograde cholangiopancreatography places a physical, psychological, and social burden on patients. In addition, endoscopic retrograde cholangiopancreatography in asymptomatic patients with stones is associated with a higher risk of general and severe complications than in symptomatic patients with stones.

According to statistical data from different studies, the overall complication rate in the asymptomatic group is higher than in the symptomatic group (26.4 % vs. 11.7 %, $P < 0.01$).[8] There is also a higher rate of severe complications in the asymptomatic group than in the symptomatic group (asymptomatic group: [16.1 %] vs. symptomatic group: [3,7 %], $P = 0{,}036$).[9]

And this question is the most difficult and debatable in solving the controversy about the mandatory cholecystectomy in patients with asymptomatic cholecystitis. Since the presence of abdominal organ pathology exists in parallel with asymptomatic cholecystitis, mimicking the clinic of exacerbations, or organ dysfunction occurs as a result of the course of cholelithiasis and, aggravated by its progression, leading to morphological changes in the intestine, pancreas, or stomach. Cholecystectomy is not indicated in patients with asymptomatic gallstones and gallbladder polyps $\leq$5 mm in size (strong recommendation based on medium-quality evidence[10]

During the 15-year follow-up asymptomatic course of intrahepatic bile duct stones was replaced by clinical manifestations

Table 1
Classification of analyzed literature sources according to the level of evidence and the degree of recommendations in accordance with the Oxford Center for Evidence-Based Medicine.

Level of evidence and Degree of recommendation	Studies encompassed
1A	Systematic review of randomized clinical trials; Systematic review of prospective cohorts; Prognostic criteria validated in several populations; Systematic review of level 1 diagnostic studies.
1B	Randomized controlled clinical trials with narrow confidence interval; Cohort from disease onset, with loss <20 %; Validated prognostic criteria in a single population; Validated cohort with a good baseline.
1C	All-or-nothing therapeutic results.
2A	Systematic review of cohort studies; Systematic review of retrospective or follow-up cohorts of untreated cases from a randomized clinical trial control group; Systematic review of diagnostic studies level >2.
2B	Cohort study (including randomized clinical trial of lower quality; e.g., <80 % follow-up); Retrospective cohort study or follow-up of untreated control patients in randomized clinical trials.
2C	Observation of therapeutic outcomes (outcomes research); Ecological study.
3A	Systematic review of case-control studies; Systematic review of diagnostic studies level >3B.
3B	Case-control study; Non-consecutive study or no consistently applied benchmarks; Non-consecutive cohort study or very limited population.
4	Case report; Case series; Low-quality cohort and case-control studies.
5	Expert opinion without explicit critical appraisal or based on physiology, bench research, or “first principles".

only in 11.5 % of patients in average in 3.4 years. The most common symptoms were colic, jaundice and fever due to cholangitis.[11] The concept of waiting therapy is also discutable, at which according to statistical data the cumulative frequency of complications from the side of biliary tracts is registered less frequently, making 6,1 % in 1 year, 11 % in 3 years and 17 % in 5 years. And asymptomatic disappearance of stones observed in 19 % of patients. When side effects associated with the procedure of early endoscopic stone removal in the asymptomatic group were observed in 32 % of patients, including 5.2 % with severe pancreatitis.[12]

However, very few academics and practicing doctors consider or choose to address this issue prior to surgical intervention, and they only start doing so following gallbladder removal when a patient's postoperative complaints in the event of asymptomatic cholecystitis continue. The persistence of symptoms after cholecystectomy is worrisome because even after successful surgery, patients need to be examined for the cause of their symptoms. This is a stressful condition even for those who believe that laparoscopic cholecystectomy is the best method of treatment of cholelithiasis. In the reviewed literature, there are no recent studies that have examined preoperative dyspepsia as a predictor factor of post-cholecystectomy syndrome. This finding could contribute to the understanding of the pathophysiology of post-cholecystectomy syndrome.[13] The causes of dyspepsia are varied, including altered gastric motility, visceral hypersensitivity, *H. pylori* infection, psychological factors, etc. The absence of gallbladder after surgery results in a continuous outflow of hepatic bile into the duodenum, which causes a rapid enterohepatic cycle of bile and excessive flow of bile juice into the duodenum. This causes increased duodenal-gastric reflux and leads to sphincter of Oddi insufficiency, which further leads to altered gastric motility. This may be the reason that patients who had preoperative dyspepsia as a result of various pathologies may develop new symptoms of dyspepsia after surgery, not as a variant of cholelithiasis.

Works on the study of asymptomatic cholecystitis are not only few, but also scattered. At the same time, there is no uniformity in both domestic and foreign literature. There is broad agreement among clinicians that patients with symptoms or complications caused by common bile duct (CBD) stones should be offered treatment for cleansing. This is strongly recommended by the best known national guidelines.[14] This consensus turns into debate and disagreement in the case of hidden CBD stones, which may be discovered during cholecystectomy with IOU or IOC, or detected on abdominal imaging performed for other reasons. Although some national guidelines recommend ductal cleansing in asymptomatic patients if they are suitable for the procedure, the evidence is of low quality and there is still no consensus on the management of these patients. These guidelines also indicate that further research on the natural history of gallbladder stone formation is still needed. Surgeons and gastroenterologists managing these patients often face an important question: do the benefits of performing endoscopic retrograde cholangiopancreatography and ductal cleansing outweigh the risks associated with the procedure? Data on the natural history of asymptomatic CBD stones are scarce. To answer this question, we need to know the natural history of asymptomatic CBD stones and what the consequences are if left alone, as well as the benefits and risks of endoscopic retrograde cholangiopancreatography in these patients.

Pancreatic pathology has been the subject of several works. The authors frequently attribute the underlying cause of glandular pathology—cholelithiasis—to the symptoms of chronic pancreatitis. They overlook the fact that pancreatic insufficiency itself may cause the body's lipid metabolism to malfunction, which may then result in an increase in the amount of lipids in bile. There is still no consensus on the association of blood lipids and bile in patients with cholelithiasis: some studies show the presence of hyperlipidemia in patients with cholelithiasis, while others deny this relationship.[15] Many studies, including epidemiologic ones, show an inverse relationship between the concentration of cholesterol and bile acids in blood serum and bile in patients with cholelithiasis. It is believed that serum lipids measured at an arbitrary moment cannot correspond to the true concentration of serum lipids at the

critical time of gallstone formation. In this case, the contradictory literature data on the causal relationship between lipid parameters and cholelithiasis are quite justified, which confirms the need for further research in this area.

A lot of people totally overlook the significance of chronic colitis in the bile recirculation process, as well as how portal blood and venous return to the liver are composed. Chronic inflammatory response changes not only the function of the reticuloendothelial system of the liver, but also the biochemical composition of bile. Causing and maintaining subacute inflammation in the biliary tract and gallbladder. There are few studies devoted to changes in gut microbiota in patients with gallstones, especially in patients with asymptomatic gallstones, and these studies have some shortcomings, for example, the influence of metabolic factors on gut microbiota is not taken into account. In a study[16] evaluating the gut microbiota between two groups, one with asymptomatic cholelithiasis and one control group, compared to healthy controls, the abundance of gut microbiota increased significantly in patients with cholelithiasis, while the diversity of microbiota decreased. The intestinal microbiota represents an important environmental factor contributing to metabolic diseases such as obesity, diabetes, dyslipidemia and hypertension,[17] all of which are closely associated with gallbladder stone formation, suggesting that the intestinal microbiota is probably related to gallstones.[18] Abnormal metabolism and secretion of cholesterol and bile acids are thought to be the primary pathophysiologic defect in gallstone formation, and the gut microbiota regulates bile acid metabolism by reducing the size and composition of the bile acid pool.[19] Many biliary diseases also depend on the gut microbiota, such as primary biliary cholangitis/cirrhosis and primary sclerosing cholangitis.[20] In another important study, the gut microbiota at different levels was shown to be altered in gallbladder stone formation, indicating that dysbiosis of the gut microbiota plays an important role in promoting gallbladder stone formation.[21] Intestinal bacteria dissociate bound bile acid into free bile acid via BSH, and then convert free bile acids into secondary bile acids via 7α-dehydroxylation. Currently, only the genus Clostridium and some strains of Lachnospiraceae and Peptostreptococcaceae in intestinal bacteria are known to possess 7α-dehydroxylation activity, and these bacteria are known as 7α-dehydroxylating bacteria. 7α-dehydroxylating bacteria have bile acid-inducible activity. The abundance and diversity of 7α-dehydroxylating bacteria in the intestinal tract of patients with gallstone disease is poorly understood, and these bacteria constitute a small fraction (<1 %) of the total gut microbiota. However, the impact of these small intestinal microbiota on the host can be enormous and they can increase the levels of secondary bile acids in the intestinal tract such as deoxycholic acid (DCA) and lithocholic acid (LCA). These secondary bile acids are returned to the liver via the enterohepatic circulation. Since 7α-hydroxylation does not occur with secondary bile acids in the human liver, bile acids can accumulate to very high levels in human bile.[22] The formation of secondary bile acids by gut microbes is of great biomedical importance. 7α-dehydroxylating bacteria convert primary bile acids to secondary bile acids, inhibiting the growth of Clostridium difficile in the body. Monocytes/macrophages, dendritic cells and natural killer T cells express the functional bile acid receptor FXR. The study revealed an increase in free bile acids and secondary bile acids in the enterohepatic circulation, which may lead to inhibition of functional FXR in immune cells. This will promote the release of inflammatory cytokines that will cause local inflammation.

Pathology of the duodenum, the most characteristic error in the indication for cholecystectomy. Pain in the epigastrium and right subcostal region is very similar to bladder colic. But for some reason surgeons have no options for additional examination of all duodenal compartments in patients with asymptomatic cholecystitis or are limited to examination of the ampulla and the duodenum. At the same time, inflammation in the duodenum, often associated with dyskinesia of the pyloroduodenal transition, and the resulting duodenostasis, lead to increased pressure in the lumen of the 12 ileum. The flow of bile from the choledochus mainly occurs under the suction effect of the 12-peritoneum or under the influence of intraductal pressure. The resulting duodenostasis leads to hypertension in the lumen of the 12-peritoneum, thereby blocking the release of bile, especially in its thickening. This is what causes the clinic of sudden onset right subcostal pain. But what is the point of vesicoureteral colic in asymptomatic cholecystitis? This remains the question for most surgeons today. In one RCT, 118 patients with pain after cholecystectomy underwent initial examination.[23] In 12 patients, microscopy revealed crystals in duodenal bile. After using ursodeoxycholic acid for several months, abdominal pain characteristic of diseases of the biliary system significantly decreased or disappeared. The data from this study confirmed that microlithiasis may be the cause of pain after cholecystectomy. Or it was initially the cause of pain syndrome, considered by clinicians as biliary colic? Then does it make sense to perform cholecystectomy when additional diagnostics to confirm microlithiasis (use microscopy of duodenal or hepatic bile obtained at endoscopic retrograde cholangiopancreatography) could exclude surgical intervention.[24] Given the relative nonspecificity of cholelithiasis symptoms, other background diseases such as functional dyspepsia, irritable bowel syndrome, (duodeno)-gastroesophageal reflux, and sphincter of Oddi dysfunction seem to be more likely causes of complaints.

Another pathology confounding surgeons is renal dysmorphia due to prolonged glomerulonephritis or pyelitis. There is some evidence that renal pathology may be a cause of post-cholecystectomy syndrome. Apparently, this question remains in the shadow of surgical community views.

5. Results

Asymptomatic cholecystitis is very common in modern clinical practice, and surgeons are observing and diagnosing an increasing number of patients due to the widespread use of abdominal ultrasound to detect abdominal diseases and unclear abdominal pain. On the one hand, some surgeons consider only patients with symptoms of typical biliary colic for surgery, since this disease is benign and complications are rare, while other surgeons, on the other hand, exaggerate any abdominal symptoms and prefer to remove any gallbladder with stones, which is facilitated by the introduction of minimally invasive cholecystectomy. The guidelines do not provide a clear management scheme for such patients, although they involve a thorough examination, they do not contain any definitive ways to treat the patient, which has led to ongoing debate between the scientific communities. Endoscopic retrograde cholangiopancreatography (ERCP), like any other treatment, is associated with certain risks. EASL recommends against treating patients with asymptomatic cholecystitis, which is a weak recommendation, since symptoms may persist even after cholecystectomy. Despite the fact that there is little data on the natural course of asymptomatic choledocholithiasis and there are no randomized controlled trials between intervention and follow-up, three national guidelines (NICE, ESGE and BSG) support surgical intervention in asymptomatic choledocholithiasis, which puts surgeons in a dilemma (Table 2).[25]

In patients with asymptomatic cholecystitis, clinical observation shows a benign clinical course. In general, patients with BC in 1–4% of cases with longer follow-up develop some concomitant symptom or complication annually (on average 6.9 years). Moreover, even if a patient with BH has a symptomatic episode, the

Table 2
Comparison of recommendations for the management of patients with asymptomatic cholecystitis.

INSTITUTE	RECOMMENDATION
NATIONAL INSTITUTE FOR HEALTH AND CARE EXCELLENCE (NICE) GUIDELINES 2014	Bile duct cleansing and cholecystectomy or people with symptomatic CBD stones.
EUROPEAN SOCIETY OF GASTROINTESTINAL ENDOSCOPY (ESGE) 2019	Recommends offering stone removal to all patients with 080 stones. with or without symptoms. who are suitable for the procedure. Strong recommendation, low-quality evidence.
BRITISH SOCIETY OF GASTROENTEROLOGY (BSG) 20I7	In line with NICE guidelines. stone removal should be offered to every patient with CBD stones. It should be noted that there are no controlled studies examining the natural history of asymptomatic CBD stones. Further research in this area is recommended.
EUROPEAN ASSOCIATION FOR THE STUDY OF THE LIVER (EASL) 2016	Symptomatic choledocholithiasis should be treated. There are no clear recommendations for the asymptomatic choledocholithiasis.

probability that this symptom will not recur is 58.5 % in mild cases and 52.1 % in moderate cases.[26] Also, in our opinion, it is important to take into account that aging may be associated with a higher risk of developing symptoms.[7] Despite the low mortality (<0.2 %) and morbidity (5–10.8 %) in laparoscopic cholecystectomy, the available data indicate that conservative treatment (clinical observation) in OCD is possible and safe. Also, follow-up without cholecystectomy is a cheaper option for the treatment of patients with BC compared with preventive cholecystectomy.[27] An inherent disadvantage of conservative treatment is that if gallstone disease is no longer asymptomatic in these patients with chronic liver disease, then the morbidity and mortality associated with cholecystectomy may be higher compared to patients without cirrhosis of the liver. However, in this group of patients, surgery and general anesthesia are considered risky [28]. While the study focuses on the cost-effectiveness of the laparoscopic method, taking into account the clinical course of asymptomatic or low-symptomatic gallstones and the risk of developing biliary colic and complications.

The second problem concerns the accidental presence of microlithiasis of the biliary tract (gallstones less than 4 mm in size, which are usually not visible during ultrasound examination of the abdominal cavity or cholecystography), which is an indication for preventive cholecystectomy. It is known that microlithiasis can cause all possible manifestations and complications of gallstone disease.[26] It is important to know that before the occurrence of any complication associated with BH (acute cholecystitis, cholangitis and acute pancreatitis), it is almost always preceded by pain, regardless of the duration of the disease.[4]

The third topic is biliary sludge. Despite the low level of recommendations and evidence, in the case of patients with low preoperative surgical risk, it can be indirectly assumed that the treatment of biliary sludge may be similar to the treatment of patients with biliary tract microlithiasis.[9] Although it has been observed that the prophylactic use of statins alone or in combination with ursodeoxycholic acid could reduce the formation of gallstones and biliary sediment.[14]

However, when used for direct therapeutic purposes to dissolve gallstones (with ursodeoxycholic acid), unsatisfactory results were obtained (dissolution of stones in 2.2 % of patients who did not take ursodeoxycholic acid). Mortality from cholelithiasis is rare, usually less than 1 %. Death from asymptomatic gallstones is extremely rare (3–7%), usually in the elderly due to biliary or postoperative complications. Symptoms develop in 10 % of patients within 5 years and in 20 % by the age of 20.[4] Autopsies show that more than 90 % of patients with cholelithiasis die from unrelated causes. It is reported that up to 40 % of patients who have undergone cholecystectomy suffer from constant pain or other abdominal symptoms – post-cholecystectomy syndrome.[5] It is necessary to pay attention to the importance of the most common complaints of patients experiencing pain associated with gallstone disease, that is, with a typical picture of pain in the biliary tract or with what has developed with a certain complication (choledocholithiasis, cholangitis, cholecystitis and pancreatitis).

Possible indications for surgery can still be strengthened, especially for those patients who present with ischemic heart disease at the same time as these conditions mentioned above, since they are more likely to develop into acute cholecystitis after 5 years of follow-up.[7]

The benefits of routine preventive cholecystectomy remain unproven, and a selective approach may be justified to avoid unnecessary risks and costs. In addition, despite the fact that laparoscopic cholecystectomy is considered a safe procedure, since the mortality rate is less than 0.2 % and the incidence is less than 5.0 %, other internal risks should be considered.[10] Unfortunately, cholecystectomy cannot guarantee the elimination of clinical manifestations detected during preoperative examination.

6. Conclusions

As a result, we would like to point out in the first paper of the series on the strategies for managing patients with asymptomatic cholecystitis that although the condition may exist, no one has made an effort to characterize it. Surgical practice, on the other hand, limits the diagnosis to the preoperative algorithm, ignoring the secondary (i.e., an outcome of another pathologic process in the organism) cause of stone formation in a patient with asymptomatic cholecystitis.

Cholecystectomy performed in this manner for asymptomatic cholecystitis merely serves to obscure and further confound the real cause of hypochondria. Thirdly, when concrements are found in the gallbladder without regard to surgical intervention, the refusal to disclose the cause of stone formation and asymptomatic cholecystitis in order to determine the course of the examination results in a number of tactical errors, maintaining the risk of post-cholecystectomy syndrome and erasing useful management recommendations for such patients. We believe that a detailed analysis of every stage of the tactical search in patients with asymptomatic cholecystitis will provide a new approach to biliary surgery and solve this problem.

Declaration of competing interest

The authors have no relevant financial or non-financial interests to disclose.

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